Muhammad Zubair Zafar

Curriculum Vitae · Software engineer and applied AI researcher

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Profile

Founding engineer and applied AI builder who works where language model agents meet systems that cannot be wrong: approval gates, deterministic checks around model output, evaluation harnesses and MCP servers that other agents call. M.S. in Computer and Information Technology (GPA 4.0, fully funded scholarship) with graduate research in deep learning for medical imaging. Ten hackathon wins, including first place at YC Haunted Hacks, MongoDB's Harness Engineering Hackathon, The Executable World, Crusoe × The AI Conference and the Amadeus × Etihad challenge. Section Leader for Stanford's Code in Place across three cycles and a long-standing volunteer instructor in data structures and algorithms.

Research Interests

- **Evaluation of agentic systems:** held-out scoring, blind and hash-locked eval tasks, LLM-as-judge with verifiable quotes, and detecting reward gaming.
- **Security and safety for AI agents:** prompt-injection path analysis, sandboxed execution, payment and refund fraud against agents that hold money, human approval gates.
- **Trustworthy multi-agent coordination:** evidence-backed agent-to-agent negotiation, voting and veto protocols, durable state and auditable receipts.
- **Physical and industrial AI:** agents that write and verify PLC programs, digital twins, on-device vision and robot learning.
- **Deep learning for medical imaging:** classification on large MRI datasets, augmentation and efficient training.

Education

M.S., Computer and Information Technology — Aug 2024 – May 2026
Eastern Illinois University — Charleston, IL, USA

- GPA 4.0 / 4.0. Fully funded master's scholarship recipient.
- Graduate Assistant software engineer (2024 – 2026) and Graduate Research Assistant in deep learning for MRI (2025 – 2026).
- Selected coursework: Advanced Database Management, Networking, Python for Data Science.

B.S., Computer Science — Oct 2019 – Sep 2023
Government College University Faisalabad, Sahiwal Campus — Sahiwal, Pakistan

- Selected coursework: Software Engineering, Artificial Intelligence.
- Won the Amadeus × Etihad challenge (2022) and completed a software engineering internship at Etihad Aviation Group while enrolled.

Intermediate in Computer Science (ICS) — 2017 – 2019
Punjab College Okara — Okara, Pakistan

Matriculation, Science Group — 2015 – 2017
Army Public School Okara Cantt — Okara, Pakistan

Research Experience

Graduate Research Assistant, Deep Learning for Medical Imaging — Aug 2025 – May 2026
Eastern Illinois University — Charleston, IL

- Trained YOLOv8 classifiers in PyTorch on more than 80,000 labeled MRI scans, reaching 97.2% test accuracy and a 95.8% F1 score.
- Built the preprocessing and augmentation pipeline (rotation, flip, scale) to improve generalization across scans.
- Moved training to Colab Pro GPUs and restructured the workflow, cutting training time by 70%.

Applied Research, Self-Improving Longevity Agent and Evaluation Harness Jun 2026 – present
Andever AI (stealth), as founding engineer San Francisco, CA

- Designed a loop in which an agent proposes interventions and an evaluation harness scores each one against 12 published epigenetic clocks.
- Held 3 clocks out of the reward on alternating rounds so that the agent could not game the metric it was optimized against.
- Paired an LLM agent (structured output, full scored history in context) with an offline optimizer that assigns credit by ridge regression, so the loop improves with or without model access.
- Cut the scoring service from 803 MB to 88 MB of memory and an eight-iteration run from 9.5 s to 0.08 s by precomputing the clock panel at build time, so it fits a 512 MB instance.

Reinforcement Learning for Humanoid Locomotion on Ice (crampon) Aug 2026
Himalaya Robotics Hack github.com/zubair480/crampon

- Trained a locomotion policy in MuJoCo that keeps a Unitree G1 humanoid upright on simulated Himalayan ice.
- Modeled ice friction, cold actuators and high-altitude wind as physical forces rather than noise.
- Achieved 2× to 6.6× longer survival on ice with no regression on normal ground.

Professional Experience

Founding Engineer Jun 2026 – present
Andever AI (stealth longevity AI startup) San Francisco, CA

- Sole engineer, working directly with the founder, investors and prospective customers. Designed and built the platform from an empty repository.
- Built the v0 engine and MCP server in TypeScript (Node 22, pnpm monorepo), validated on mock patient data. It ingests lab PDFs, CGM exports and wearable data, flags abnormal biomarkers, groups them by aging mechanism and drafts a protocol in which every item cites graded evidence.
- Split the system into a core engine, a provider-agnostic LLM layer with a deterministic fallback, and an MCP server over stdio and streamable HTTP. The deterministic and LLM paths share the same ranking, citation and safety logic.
- Enforced a clinician approval gate in the core engine rather than the UI, so no transport can release unapproved output. Dangerous lab values always produce a referral and contraindicated items are blocked.
- Implemented an AES-256-GCM encrypted store with key rotation and delete-on-request, plus bearer tokens or OAuth 2.1 (PKCE, dynamic client registration) on the HTTP transport.
- Shipped with 787 tests and a blind evaluation that passed 34 of 34 hash-locked tasks. Fuzzing with 1,240 invalid inputs per transport caused zero crashes, and p95 latency is under 500 ms on every tool. CI runs lint, typecheck, tests, build and evals.

Software Engineer (Graduate Assistant) Aug 2024 – May 2026
Eastern Illinois University Charleston, IL

- Shipped an enterprise HR system in Laravel for more than 7,000 users with SSO and role-based access, and cut API response time by 65% through query tuning and indexing.
- Rewrote a 2006-era PHP application from the ground up in Laravel. Reliability rose 35% after the migration.
- Automated request workflows with Twilio SMS and REST APIs, reducing manual data entry by 40% and request processing time by 25%.
- Containerized services with Docker and worked with systems teams on dependable CI/CD deployments.

Backend Software Engineer Dec 2023 – Aug 2024
WPBrigade

- Developed and maintained more than 5 production WordPress plugins with over 15,000 combined active installs, including custom REST endpoints, webhook integrations and database schema optimization.
- Wrote a modular PHP and MySQL theme framework that lets more than 200 non-technical users build sites without code. Deployment time fell 60%.
- Led code reviews for a team of five, mentored three junior developers and delivered secure coding bootcamps that reached more than 100 students.

Software Engineering Intern, Cargo Sep 2023 – Oct 2023
Etihad Aviation Group, World Headquarters Abu Dhabi, UAE

- Turned the hackathon-winning IATA cargo message (CPM) parser into a production tool (Python, Flask, regular expressions). Manual processing fell 30% and data-entry errors 95%.
- Replaced paper workflows with Power Apps and Power Automate, improving cargo data accuracy by 40%.
- Combined Timatic APIs with spaCy to verify travel requirements for more than 200 countries in real time.
- Created a developer tool that generates UML diagrams from Python codebases, used on more than 15 internal projects.

Software Engineer (MERN stack) 2022 – 2023
CarChaze

- Built the marketplace experience and core APIs for a vehicle platform serving more than 2,000 monthly users, with MongoDB, Express, React and Node.js.
- Engineered more than 10 core REST endpoints and improved data processing performance by 15%.
- Managed Firebase and Heroku deployments through continuous integration and delivery.

React Developer May 2022 – Jul 2022
Koboldo

- Developed product features in React.

Honors and Awards

Hackathon wins (10)

Oct 2026	**1st place**	YC Haunted Hacks, San Francisco, for *ActorRelay* (team).
Oct 2026	**1st place**	MongoDB Harness Engineering & Model Wrangling Hackathon, SF Tech Week, Shack15, San Francisco, for *MongoMation* (team).
Oct 2026	**3rd place**	"Build an Agent" Hackathon (Agent37, InsForge and Monid), Corgi Cafe, San Francisco, for *PlantAPI* (team). $1,000 cash and $1,000 in OpenAI credits.
Sep 2026	**1st place**	Crusoe × The AI Conference hack day, San Francisco, for *Thermal Crusoe* (team of 4). $5,000 first prize.
Sep 2026	**1st place**	The Executable World hackathon, San Francisco, for *RefundGuard*.
Sep 2026	**2nd place + 2 tracks**	B.E.L.L.E × Qoder × Neo4j hackathon, San Francisco, for *RecallRadius* (team of 3). Won the Neo4j prize and the Qoder prize.
Aug 2026	**2nd place**	Agent Native Builders Hackathon, Cloudflare HQ, San Francisco, for *Attest* (solo build).
Jul 2026	**2nd place**	Auth0 × Stripe Hackathon, Okta HQ, San Francisco, for *Clip Police* (team).
Jul 2026	**2nd place**	CrewAI Hackathon, San Francisco, for *ROOT* (team).
2022	**1st place**	Etihad challenge, Amadeus Worldwide Hackathon (Etihad Green Development Challenge), for *Smart CPM Parser* (team). The win led to an internship at Etihad, where the parser went into production.

Scholarships, competitions and other recognition

2024 – 2026	**Scholarship**	Fully funded master's scholarship, Eastern Illinois University.
2023 – 2025	**Selection**	Selected from thousands of applicants as a Section Leader for Stanford's Code in Place, three cycles.
2023	**Award**	Transatlantic AI Hackathon: awarded an OAK-D camera for *Smart Safety*, a real-time PPE detection system.
2023	**National top 20**	Meta Hacker Cup: 16th nationally.
2023	**Team lead**	Harvard CS50 Puzzle Day: led a top-performing team.
2022	**25 of 25 days**	Advent of Code: completed the full 2022 challenge.
	Worldwide rank	Google Code Jam: placed 1,622 worldwide.
Ongoing	**460+ problems**	LeetCode: more than 460 problems solved.

Teaching and Mentoring

Section Leader, Code in Place (CS106A material) 2023 – 2025, three cycles
Stanford University (online) International cohort

- Taught Python and problem solving to an international section of learners, selected from thousands of volunteer applicants.
- Ran weekly section meetings with interactive discussion and answered questions to strengthen understanding of core programming concepts.
- Reviewed assignments and projects with constructive feedback on correctness and good practice.
- Organized collaborative coding sessions and acted as the liaison between students and course staff.

Data Structures and Algorithms Instructor and Senior Moderator 2022 – present
iCodeGuru (volunteer community for Pakistani students) Online

- Moderated and taught the data structures and algorithms sessions, guiding participants through LeetCode problems.
- Organized a weekly webinar series with accomplished professionals giving technical talks.
- Selected and prepared students for high-quality hackathons.
- Taught IELTS classes for two months to students who could not afford preparation courses.

Business and Technical Mentor 2023 – 2024
lablab.ai hackathons Online

- Facilitated team formation, guided teams in accessing APIs and integrating external services, and coached pitching.
- Supported teams through technical blockers to timely submissions, alongside other mentors.

Secure Coding Bootcamps and Developer Mentoring 2023 – 2024
WPBrigade

- Delivered secure coding bootcamps reaching more than 100 students and mentored three junior developers.

Service and Leadership

- **Organizer**, WordSprint Developers Hackathon, 2024.
- **Mentor**, lablab.ai hackathons, 2023 – 2024.
- **Senior Moderator**, iCodeGuru community, 2022 – present.
- **Scholarship guidance**: recorded a webinar and wrote a step-by-step guide for students applying to fully funded MS and PhD programs in the US, sharing the CV and professor outreach email used in his own application, and answers applicants' questions directly.

Research Projects and Systems

Detailed entries for the work closest to research. Every project in the portfolio is listed in the next section.

ActorRelay · 1st place, YC Haunted Hacks (team) Oct 2026

An agentic control layer on a simulated two-plant process unit. Every machine has a durable actor holding its memory and its own AI agent; a setpoint changes only if every connected machine votes yes while citing past runs, and a human has a 10-second veto. One refusal blocks the change with its evidence, the pump's agent offers a safer target, and a rule learned in one plant is cited in the other.

Evaluation: 26 of 26 full-stack end-to-end checks on real actors, a real relay and real model calls. Crash testing with `kill -9` passed 32 of 33 scenarios with no acknowledged write lost across 37 server kills. **Stack:** Terse durable actors, Agent Relay, OpenAI models, TypeScript, Bun, three.js.

MongoMation · 1st place, MongoDB Harness Engineering Hackathon (team) Oct 2026

A self-improving autonomous PLC engineer. An agent turns a plain-English machine specification into an IEC 61131-3 Structured Text program, compiles and deploys it to a real OpenPLC runtime, and tests it against a live Factory I/O plant on AWS. An independent grader with hidden tests decides pass or fail and drives a repair loop, while a meta-agent proposes harness changes behind a benchmark-gated keep or rollback step.

Results: best custom specification met 8 of 8 requirements and passed 14 of 14 public and hidden tests after one repair; 7 fully verified projects and 473 graded tests; a repeated specification fell from 290 s and $0.073 to 1 s and $0 by recalling a rig-proven program. **Stack:** MongoDB Atlas (change streams, transactions, hybrid vector and text search), Voyage AI, Claude, Raindrop, Modelence, OpenPLC, Factory I/O, TypeScript.

PlantAPI · 3rd place, Build an Agent Hackathon (team, with Ali Amjad) Oct 2026

An eleven-agent maintenance team for an industrial plant, all running on Agent37. An operator uploads a photo of a failure; agents triage it, check repair history, source the part, find an outage window and a qualified technician, resolve disagreements into one plan, tag every action for risk and lockout/tagout, and after one human approval open the work order, email the supplier, book the technician and verify the completion photo.

Result: photo to closed work order in about 11 minutes on real systems for $0.13 of agent compute. **Stack:** Agent37, OpenAI vision, Supabase, Monid, InstaCloud, Fiix, Odoo, Next.js.

RefundGuard · 1st place, The Executable World Sep 2026

A release gate for AI agents that hold a wallet. Eleven adaptive scam personas attack a refund agent in a simulated store while three honest customers act as the control. Every ledger movement is replayed against policy in code, and an LLM judge's findings count only when they quote the agent verbatim.

Result: version 1 leaked $310 and promised $709 in writing; by version 3, 0 of 11 attacks breached and every honest customer was served. **Stack:** Node.js, LLM judge, ledger replay, server-sent events.

Thermal Crusoe · 1st place ($5,000), Crusoe × The AI Conference (team of 4) Sep 2026

A 3D thermal twin of a motor control center room in the browser. Clicking a bucket zooms from line-up to contactor, where a seeded simulation drives per-phase current and terminal temperatures, one joint runs hot, and the app drafts the inspection report. Built the three.js twin and owned the Next.js APIs, shared contracts and integration.

Evaluation: a 180-request evaluation of the Crusoe-hosted model passed every check at about 681 ms median latency. **Stack:** three.js, Next.js, Crusoe, Plaud.

Attest · 2nd place, Agent Native Builders Hackathon (solo) Aug 2026

Agent-to-agent vendor security review in place of static compliance questionnaires. A bank's agent questions a vendor's agent, which runs with separate credentials. Answers count only when evidence backs them: a deterministic severity counter and an LLM relevance judge must both agree, disagreements escalate to a person, and every decision leaves a receipt with the SHA-256 of its raw evidence.

Stack: Cotal.ai agents, Runtype.ai, JavaScript; four evidence collectors with judgments cached on the evidence digest.

RecallRadius · 2nd place + Neo4j and Qoder track prizes (team of 3) Sep 2026

A vehicle part traceability workspace that connects supplier lots, affected vehicles, distributors and previous repairs into one Neo4j graph an engineer can follow. Owned the contracts, APIs and integration tests.

Result: 114 tests and 23 of 23 acceptance checks against Neo4j Aura. **Stack:** Next.js, TypeScript, Neo4j.

PayeeLock · AI Security Hackathon SF Sep 2026

A defense for an AI accounts-payable agent against business email compromise. Agent-written code runs in a Wasmer sandbox with the network switched off, and a ledger that never reads email decides payments.

Result: the demo paid the legitimate $48,250 invoice and blocked the spoofed one. **Stack:** Node.js, Wasmer, AgentMail, Express.

Redline · Prompt-injection scanner Jul 2026

Models an agent's tools as a graph and enumerates paths from untrusted input to a privileged action that cross no guard. Built the scan pipeline, row-level-security authentication and the fix endpoint.

Stack: TypeScript, static analysis, graph search.

JudgeAgent · Agent as Judge (contributor) Jun 2026

A verification layer for multi-agent handoffs: it derives a rubric from the goal, checks whether an agent's claim is actually proven by its evidence, and returns a traceable verdict before the next agent trusts it.

Stack: Python, MCP server with a workbench and consensus mode, W&B Weave, Next.js.

MCP Auditor · Harness Engineering Hack (contributor) Jun 2026

Six parallel agents probe live MCP servers against the SAFE-MCP catalog for path traversal, leaks, tool poisoning and exfiltration, and publish citation-grounded reports. A clean negative control must return zero findings.

Stack: multi-agent, SAFE-MCP, ClickHouse, x402 settlement.

EstateAgent AI · HackIllinois 2026 Mar 2026

A map-reduce agent pipeline on Modal with parallel analyst agents (Qwen 2.5 on vLLM, H100 GPUs) and FastAPI that scores real estate investments, streams progress over WebSockets on Cloudflare Durable Objects and runs Monte Carlo pricing simulations at the edge.

Result: 5,000 simulations in under 12 seconds; live demo. **Stack:** FastAPI, Modal, vLLM, Cloudflare.

Complete Project Portfolio

Thirty-four builds listed on zubair480.github.io/projects, plus related repositories. Newest first within each group.

Agentic AI, evaluation and security

Faceplate · Own Your Intelligence (YC) Sep 2026

An AI copilot for industrial HMI screens: click a pump or alarm table on a live Ignition screen, describe the change in plain English, and it proposes the edit in the team's house style using a model the plant owns (Qwen LoRA, River, GBrain, TypeScript). Code · demo.

PLVA · GPT-6 Astra Hackathon Sep 2026

Private AI assistance for client work: the assistant plans against protected screenshots and opaque references while real client values stay behind a local boundary, and receipts show exactly what reached the provider (Python, Astra, browser agent, local inference). Code.

S0101 · Intel Labs Physical AI track, AI Infra Summit Sep 2026

Robot manipulation and anomaly detection on Intel Labs Physical AI Studio, from dataset collection to evaluation, with two ACT models plus SmolVLA and an Anomalib safety net.

AgenticAudit · Web Data UNLOCKED (backend contributor) Jul 2026

A proactive compliance agent that scans websites, collects screenshot evidence and flags risks across GDPR, SOC 2, HIPAA and PCI through an asynchronous evidence pipeline from scrape to alert (FastAPI, Bright Data, RAG, Pydantic). Code.

Clip Police · 2nd place, Auth0 × Stripe Hackathon Jul 2026

A multi-agent creator-protection platform that finds unauthorized reposts of creator videos across the web and assembles a real-time circulation report (multi-agent, Auth0, Stripe, content detection).

ROOT · 2nd place, CrewAI Hackathon Jul 2026

Fourteen autonomous agents coordinate employee onboarding from a single Zendesk request, provisioning software, hardware and access and compressing a week of work into minutes (CrewAI, Zendesk).

Ops Sentinel Jul 2026

An autonomous incident response agent that grounds findings in live search, then dispatches guarded rollbacks (JavaScript). Code.

AI GTM Agent · AWS Builder Loft Jul 2026

An autonomous feedback loop that listens across social channels, verifies reported product issues, fixes authorized problems and responds with evidence; a judge volunteered to become the first customer (multi-agent, Nimble, Kylon).

Thread · Context over Amnesia Jun 2026

An offline-first research companion that turns a paper into a structured summary with a methodology graph and grounded conversation using local models (FastAPI, Ollama, React, Mermaid). Code.

Model Compatibility Agent · Agentic AI Hackathon Jun 2026

An agent pipeline that extracts a deep learning model's operations, checks them against Vitis AI documentation with retrieval, and proposes a compatible architecture using only supported functions; runs as a CLI, Streamlit app or messaging backend (Python, RocketRide, XTrace). Code.

Industrial, physical and mobile AI

PEMA · Physical AI, AI Infra Summit (team of four) Sep 2026

On-device environmental memory for low-vision and memory-impaired users that remembers where objects were last seen and speaks up when something moves, running on the Snapdragon NPU with an STM32 confirming physical actions, with no cloud and no face data.

Postcard · Bitrig Hacks, iPhone Duo at YC Sep 2026

A photo and handwritten feeling note sent to one person, where unfolding the device opens the card and folding seals it, with sending kept as a separate tap (SwiftUI, Supabase, iOS). Code.

Electric Lens · Qualcomm × Meta Hackathon Jun 2026

An offline, camera-guided safety assistant for electricians that verifies lockout and tagout steps and produces an evidence-backed permit, with private zero-internet inference on device (Kotlin, ExecuTorch, CameraX, QNN). Code.

Hifz · personal project Jul 2026

A Qur'an memorization companion with SM-2 spaced repetition, offline reciter audio, word-level highlighting and recitation scoring (Kotlin, Jetpack Compose). Code.

Smart Safety · Transatlantic AI Hackathon Jul 2023

A real-time workplace safety system that detects protective equipment with an OAK-D camera and alerts supervisors with employee-aware notifications (YOLOv8, OAK-D, Python, Flask). Code.

Navi Tutor · lablab.ai OpenAI Hackathon 2023
An AI learning platform that retrieves from curated articles and uses language models to answer questions across subjects (Python, Haystack, Elasticsearch, OpenAI).

Codic · OpenAI Hackathon Dec 2022
A voice-led creative app that turns spoken prompts into generated images through a mobile interface and an AI-powered Django backend (Whisper, OpenAI API, Django, React Native). Code.

Products, data and developer tools

Party Line · Spatial Intelligence and Generative 3D Hackathon Sep 2026
Two players stand in the same kitchen twenty-eight years apart, and what the 1998 player does rewrites the 2026 player's world; one Convex transaction writes both timelines at once (Convex, three.js, Tripo). Code.

Successor · MongoDB hackathon Aug 2026
Captures a retiring engineer's troubleshooting knowledge; owned retrieval (Atlas Search and Vector Search with rank fusion) and a LangGraph executor checkpointed to MongoDB. Code.

Tenderly · AI for Social Good (backend contributor) Jul 2026
An AI volunteer-matching platform for San Francisco residents and real nonprofits, using live SF 311 data to surface urgent community needs (FastAPI, DigitalOcean AI, React). Code.

Signal SF · personal project Jul 2026
A personalized San Francisco events finder with multi-user profiles, itinerary planning and conflict warnings (Node.js, SQLite, Vercel). Code.

FlashCard · RunPod Flash Hack Day Jun 2026
A QR-powered networking experience that turns a LinkedIn profile into a personalized trading card and lead signal, with GPU generation and a live scan dashboard (FastAPI, RunPod, Bright Data, Flux). Code.

Micro Loan Negotiator · Gen UI Hack Jun 2026
A mobile-first lending experience whose generative interface lets an agent choose which cards and charts to render from a widget registry at runtime (Flutter, Dart, Supabase). Code.

Smart CPM Parser · 1st place, Amadeus × Etihad 2022 – 2023
A cargo message parser that validates and corrects operational CPM messages before they reach production workflows; the hackathon MVP later went into production at Etihad with 95% fewer entry errors and 30% less manual work (Python, Flask, regular expressions, JavaScript). Code.

zubair480.github.io · personal site 2026
This portfolio: a Next.js 16 static export on GitHub Pages with CSS motion and progressive WebGL enhancement, server-rendered tests and an automated build.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, PHP, Kotlin, Dart, Swift, C.

Agents and LLMs: MCP (stdio and streamable HTTP), LangGraph, CrewAI, evaluation harnesses, LLM-as-judge, RAG, vLLM, OpenAI and Anthropic APIs.

Machine learning: PyTorch, YOLOv8, scikit-learn, MuJoCo, ExecuTorch, NumPy, Pandas, TensorFlow.

Backend and web: Node.js, FastAPI, Flask, Django, Laravel, Express, Next.js, React, React Native, REST, WebSockets.

Data: PostgreSQL, MongoDB Atlas (vector and text search), Neo4j, MySQL, SQLite, Redis, Supabase, Convex, ClickHouse.

Infrastructure: Docker, GitHub Actions and CI/CD, Cloudflare Workers and Durable Objects, Modal GPUs, AWS EC2, Render, Wasmer sandboxing.

Industrial: OpenPLC, IEC 61131-3 Structured Text, Factory I/O, CODESYS, Ignition.

Spoken languages: Urdu (native), English (fluent).

References

Available on request.